WORKED SOLUTION

STEP I 2004, Question 6

Pure • Vector Geometry of Triangles

STEP I 2004, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q6

The key idea

Work with position vectors rather than separate x- and y-coordinates: write $\mathbf{a} = (p_1, q_1)$, $\mathbf{b} = (p_2, q_2)$, $\mathbf{c} = (p_3, q_3)$, so that H has position vector $\mathbf{a} + \mathbf{b} + \mathbf{c}$. Every line in the question then has a short vector description, and "at right angles" becomes "the scalar product is zero". The scalar product that turns up is a difference of two squares, $(\mathbf{c} + \mathbf{b}) \cdot (\mathbf{c} - \mathbf{b})$, and that is the whole second half.

Solution

We assume A, B, C are not collinear, so that they form a genuine triangle (otherwise the lines in the first part can coincide and there is no single point of intersection).

The medians meet

Let L, M, N be the midpoints of BC, CA, AB, with position vectors

$$\mathbf{l} = \tfrac{1}{2}(\mathbf{b} + \mathbf{c}), \qquad \mathbf{m} = \tfrac{1}{2}(\mathbf{a} + \mathbf{c}), \qquad \mathbf{n} = \tfrac{1}{2}(\mathbf{a} + \mathbf{b}).$$

A general point of the line AL is

$$\mathbf{a} + s(\mathbf{l} - \mathbf{a}) = (1 - s)\,\mathbf{a} + \tfrac{s}{2}\,\mathbf{b} + \tfrac{s}{2}\,\mathbf{c},$$

and a general point of the line BM is

$$\mathbf{b} + t(\mathbf{m} - \mathbf{b}) = \tfrac{t}{2}\,\mathbf{a} + (1 - t)\,\mathbf{b} + \tfrac{t}{2}\,\mathbf{c}.$$

These two expressions agree if the coefficients of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ match: $1 - s = \frac{t}{2}$, $\frac{s}{2} = 1 - t$ and $\frac{s}{2} = \frac{t}{2}$. The last gives $s = t$, and then $1 - s = \frac{s}{2}$ gives $s = t = \frac{2}{3}$, which satisfies all three. So both lines pass through the point G with position vector

$$\mathbf{g} = \tfrac{1}{3}(\mathbf{a} + \mathbf{b} + \mathbf{c}), \qquad \text{that is,} \qquad G = \left(\tfrac{p_1+p_2+p_3}{3},\ \tfrac{q_1+q_2+q_3}{3}\right).$$

Ask yourself Matching coefficients found *a* common point. How do we know it is *the* point of intersection?

Two distinct lines meet in at most one point, so we only need AL and BM to be different lines. If they were the same line, it would contain A, B and L, hence the whole line BL, which is the line BC; then A would lie on BC, contradicting our assumption. So G is the unique point of intersection.

For the third median CN, take the point $\mathbf{c} + u(\mathbf{n} - \mathbf{c})$ with $u = \frac{2}{3}$:

$$\mathbf{c} + \tfrac{2}{3}\left(\tfrac{1}{2}\mathbf{a} + \tfrac{1}{2}\mathbf{b} - \mathbf{c}\right) = \tfrac{1}{3}\mathbf{a} + \tfrac{1}{3}\mathbf{b} + \tfrac{1}{3}\mathbf{c} = \mathbf{g}.$$

So G lies on the line joining C to the midpoint of AB, as required. (The value $\frac{2}{3}$ each time says that G, the centroid, is two-thirds of the way along each median from the vertex.)

Perpendicularity at H

The point H has position vector $\mathbf{h} = \mathbf{a} + \mathbf{b} + \mathbf{c}$, so

$$\overrightarrow{AH} = \mathbf{h} - \mathbf{a} = \mathbf{b} + \mathbf{c}, \qquad \overrightarrow{BC} = \mathbf{c} - \mathbf{b}.$$

If the line AH meets BC at right angles then these direction vectors have zero scalar product:

$$0 = (\mathbf{c} + \mathbf{b}) \cdot (\mathbf{c} - \mathbf{b}) = \mathbf{c} \cdot \mathbf{c} - \mathbf{b} \cdot \mathbf{b} = (p_3^2 + q_3^2) - (p_2^2 + q_2^2),$$

where the cross terms $\mathbf{c} \cdot \mathbf{b}$ and $\mathbf{b} \cdot \mathbf{c}$ cancel because the scalar product is commutative. Hence

$$p_2^2 + q_2^2 = p_3^2 + q_3^2,$$

which is the required result. In words, B and C are the same distance from the origin O.

In exactly the same way, $\overrightarrow{BH} = \mathbf{a} + \mathbf{c}$ and $\overrightarrow{AC} = \mathbf{c} - \mathbf{a}$, so if BH meets AC at right angles then $(\mathbf{c} + \mathbf{a}) \cdot (\mathbf{c} - \mathbf{a}) = 0$, giving

$$p_1^2 + q_1^2 = p_3^2 + q_3^2.$$

The third altitude

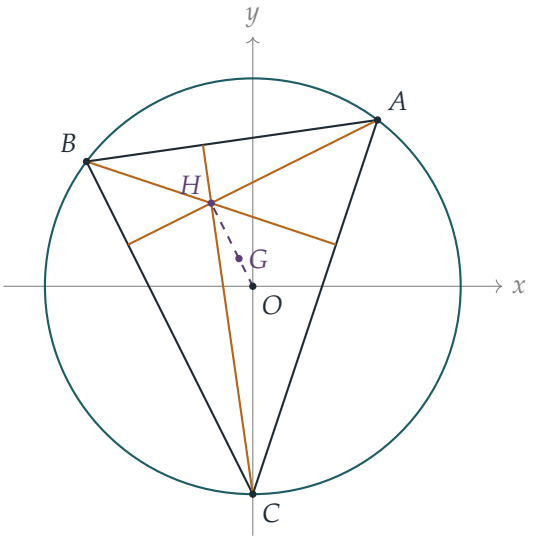
Suppose both perpendicularity conditions hold. Combining the two results above,

$$p_1^2 + q_1^2 = p_3^2 + q_3^2 = p_2^2 + q_2^2, \qquad \text{so} \qquad \mathbf{a} \cdot \mathbf{a} = \mathbf{b} \cdot \mathbf{b}.$$

Now $\overrightarrow{CH} = \mathbf{a} + \mathbf{b}$ and $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$, and

$$\overrightarrow{CH} \cdot \overrightarrow{AB} = (\mathbf{b} + \mathbf{a}) \cdot (\mathbf{b} - \mathbf{a}) = \mathbf{b} \cdot \mathbf{b} - \mathbf{a} \cdot \mathbf{a} = 0.$$

So CH is perpendicular to AB (provided $H \neq C$, so that CH is a genuine line; if $H = C$ the statement is empty). This completes the deduction.



The figure shows the situation in the final part: A, B, C lie on a circle centred at the origin O, the three altitudes (amber) all pass through H, and O, G, H are collinear with $\mathbf{h} = 3\mathbf{g}$.

Checks

- Take $A = (3,4)$, $B = (-4,3)$, $C = (0,-5)$, all at distance 5 from O. Then $H = (-1,2)$, and $\overrightarrow{AH} \cdot \overrightarrow{BC} = (-4)(4) + (-2)(-8) = 0$; the other two scalar products are also 0. The centroid is $G = (-\frac{1}{3}, \frac{2}{3})$, one third of H.
- Numerically, for about 2000 random triangles with integer vertices, the intersection of two medians computed exactly with fractions was always $\frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$ and lay on the third median; and for 1000 random triangles inscribed in a circle centred at O, all three scalar products for H vanished.

Common slips

- Showing the two medians pass through G without saying why they cannot be the same line. One sentence about A, B, C not being collinear settles it.
- Using $\overrightarrow{OH} = \mathbf{a}+\mathbf{b}+\mathbf{c}$ as the direction of AH. The direction is $\mathbf{h}-\mathbf{a} = \mathbf{b}+\mathbf{c}$.
- Thinking the result holds for every triangle. The hypothesis forces O to be equidistant from A, B, C; for a general origin, H is not the orthocentre.

Beyond STEP

The final part proves that the three altitudes of a triangle are concurrent: place the origin at the circumcentre O (equidistant from the vertices), and the point $\mathbf{a}+\mathbf{b}+\mathbf{c}$ lies on all three. Since $\mathbf{h}=3\mathbf{g}$, it also proves that the circumcentre, centroid and orthocentre lie on a line, the *Euler line*, with $OG:GH=1:2$. This vector style of geometry, where coordinate-free identities do the work, is how geometry is done in first-year linear algebra; Strang's *Introduction to Linear Algebra* develops it, and Siklos's *Advanced Problems in Mathematics* has more STEP-style practice.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.